

A novel multiplex PCR assay for individual identification in the vulnerable Chinese Egret (*Egretta eulophotes*) using molted feathers

Yufei Dai¹ · Wenzhen Fang^{1,2} · Qingxian Lin^{1,2} · Xiaoping Zhou^{1,2} · Xiaolin Chen^{1,2}

Received: 13 November 2014 / Accepted: 1 April 2015 / Published online: 22 April 2015
© Springer Science+Business Media Dordrecht 2015

Abstract Noninvasive DNA may be used in individual identification for counting or tracking animals. Here a multiplex PCR assay, in which 16 highly polymorphic microsatellite markers were chosen and allocated into four multiplex PCR panels, was described for the individual identification of the Chinese Egret (*Egretta eulophotes*) using molted feathers. In this study, the combined exclusion probability for unrelated individuals (P_{ID}) and for individuals with sibling relationship ($P_{ID(sib)}$) were 7.02×10^{-19} and 2.9×10^{-7} , respectively. Molted feathers ($n = 59$) of *E. eulophotes* amplified by more than 7 loci were unambiguously identified as 43 different individuals (32 females and 11 males). The results indicated that our newly developed assay could be used for noninvasive individual identification in the vulnerable Chinese Egret and other Ardeids.

Keywords Multiplex PCR · Microsatellite genotyping · Individual identification · Noninvasive DNA · Egret

Electronic supplementary material The online version of this article (doi:10.1007/s12686-015-0464-0) contains supplementary material, which is available to authorized users.

✉ Xiaoping Zhou
xpzhou@xmu.edu.cn

✉ Xiaolin Chen
xlchen@xmu.edu.cn

¹ Key Laboratory of Ministry of Education for Coast and Wetland Ecosystems, School of Life Sciences, Xiamen University, Xiamen 361102, People's Republic of China

² Present Address: Key Laboratory of Ministry of Education for Coast and Wetland Ecosystems, College of the Environment and Ecology, Xiamen University, Xiamen 361102, People's Republic of China

Microsatellite genotyping can be used to identify individuals and thus count or track animals. For endangered birds, individual identification studies are rare because of difficulties associated with capturing them and obtaining samples (Rudnick et al. 2005). Recently, many studies have demonstrated that molted feathers can provide an easily available source of DNA in avian studies (Horvath et al. 2005; Isabel Miño and Nassif Del Lama 2009). However, low-quality DNA in molted feathers may lead to low amplification success rates in polymerase chain reaction (PCR) or failure of locus amplification in microsatellite genotyping (Johansson et al. 2012). In this study, a novel multiplex PCR assay, based on highly polymorphic microsatellite markers, was developed for individual identification of molted feathers in the vulnerable Chinese Egret (*Egretta eulophotes*) (IUCN 2014).

Sampling, storage and DNA isolation of molted feathers and plucked nestlings' feathers were described in the supplementary material. The DNA samples were amplified using species-specific ND6 and tRNA^{Glu} gene primers for species identification (Huang et al. 2012) and using Chromo-Helicase-DNA binding protein gene (CHD) primers for sex determination (Wang et al. 2011). Then, 13 microsatellite loci isolated from the Chinese Egret (Huang et al. 2010; Dai et al. 2013) and three markers obtained by cross-amplification from Reddish Egret (*Egretta rufescens*) (Hill and Green 2011) were chosen and allocated into four multiplex PCR panels using software MPprimer (Shen et al. 2010), based on size limitations and annealing temperature of primers. The 5' end of each forward primer was labeled with different fluorescent dyes (HEX, TAMRA or FAM), and the Multiplex PCR panels were performed using TaKaRa Multiplex PCR Assay Kit in 20 µL reaction volume containing 10 µL of Multiplex PCR Mix 2, standardly diluted primer mix, 0.1 µL of Multiplex PCR Mix 1,

Table 1 Sixteen microsatellite loci arranged into four multiplex PCR panels for the Chinese Egret (*Egretta eulophotes*)

Multiplex panel	Locus	GenBank accession	Primer sequence (5'–3')	T _{multi} (°C)	Fluorescent dyes	Size range (bp)	k	N	Ho	He	PIC	HWE
I	Ae36	FJ900150	F: TGCTGAGTAAACAGGCAGTTGGGT R: TCCCCCTGCAGAAAGGATGTGGT	62	FAM	94–114	6	134	0.791	0.816	0.785	0.283
	Ae01	FJ900138	F: TGAGCTCTTGCTGGCATCCACCT R: ACACCTTCTCCACCAGCAGCTTGT		TAMRA	142–162	6	134	0.552	0.646	0.580	0.049
	Ee28	KJ748387	F: CGCCAGCTTGTTAATGCTACAAG R: CAGAGACACAGACAGGAAGAGAG		HEX	118–139	8	134	0.694	0.693	0.641	0.521
	Ee46	KJ748397	F: GAAACAGAGAGACTTCCAAACCCTG R: TGAGGCTCTGGTGATGTTGGTG		FAM	172–200	8	133	0.586	0.746	0.720	0.009
	Ae44	FJ900154	F: AGGGAGTCATGGTGGGATAATTCA R: CATCTTCTGGCCCTGTCTTCT	62	FAM	106–122	5	133	0.684	0.634	0.562	0.431
	Ae37	FJ900151	F: CCAGTACAACTCCACGGAGAGCTG R: CTACTTTAAAGATGACCGGACAGGT		TAMRA	165–213	8	134	0.612	0.636	0.600	0.034
II	Ae50	JF433925	F: AGCCTTTCTGTCTCTGCCCCTCA R: AGCAGGGTTGCATCTCCAGCAC		FAM	210–262	12	134	0.858	0.823	0.797	0.013
	Ee45	KJ748396	F: ACCGAAGTTGGAAGGAAGGGAAG R: TAGCATCACAGTGGTGTGGAG		HEX	125–165	8	134	0.716	0.697	0.654	0.462
	Ee46	HM368083	F: AGGGAAGAAAGAGAGGGACA R: CTGTTTGCTAATTTATTTGGCATCTATC	56	FAM	117–205	17	133	0.812	0.852	0.831	0.076
	Ae05	FJ900140	F: AGACCGGGTTTATTTTGGCCTT R: CGGGCATTTTCTTCTTCTGACGC		TAMRA	177–201	7	132	0.879	0.813	0.782	0.005
	Ae52	JF433927	F: TGCTGCAAGTAAAGACAACGGA R: TGGGATGAACCTCATATTAGACTTGG		FAM	193–438	32	133	0.955	0.955	0.949	*
	Ee04	KJ748377	F: TAGGAGATGGGACGGCAGGGAC R: CAGCACCTGAAGCCAGACGATC		HEX	133–183	10	132	0.788	0.834	0.811	0.344
IV	Ae51	JF433926	F: AGCTGGGCACTTAGTTATCCCA R: TGCTCCCAAAAGACTGAGGACC	56	FAM	110–146	7	134	0.597	0.633	0.562	0.380
	Ee51	HM368079	F: GGCCTGCGCACTTAAGTCAAGT R: CTCAGGCAGTTGGACTAGATGA		TAMRA	123–208	15	134	0.933	0.906	0.895	0.750
	Ee44	HM368086	F: CAAAGCAACAAACATTCAGCAATT R: CGAGAACTTGAGGGCTGGCT		FAM	210–238	8	134	0.687	0.695	0.641	0.051
	Ee18	KJ748381	F: GCAGTTGGACTAGATGAACATTGT R: CTTCCATGATAAACCAATACCCAGC		HEX	171–216	10	134	0.843	0.818	0.790	0.037

T_{multi} (°C), annealing temperature used for multiplex PCR; K, number of alleles; N, number of individuals analyzed; Ho, observed heterozygosity; He, expected heterozygosity; PIC, polymorphic information content; HWE, probability of the exact test for deviation from Hardy–Weinberg Equilibrium

* Significant deviation from HWE after Bonferroni correction

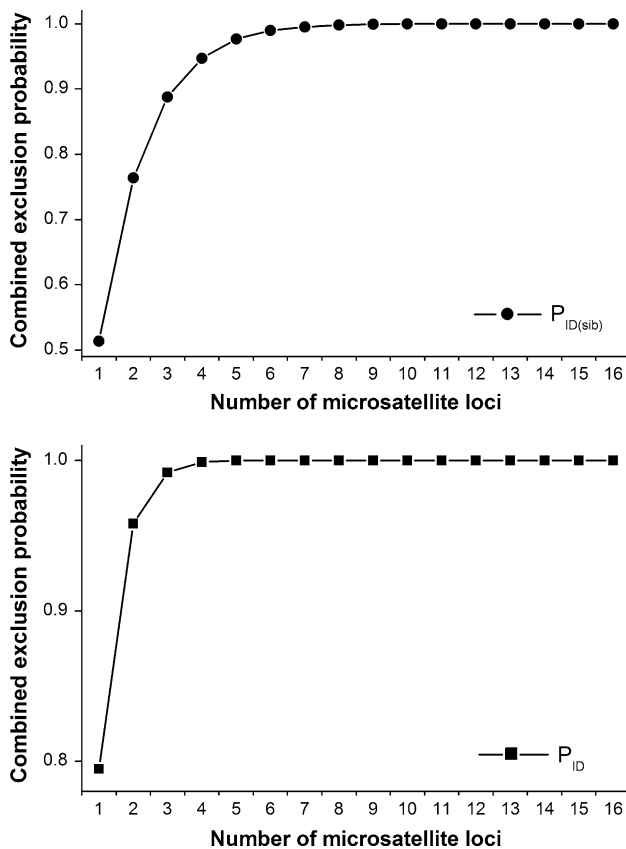


Fig. 1 The combined exclusion probability for unrelated individuals (P_{ID}) and for individuals with sibling relationship ($P_{ID(sib)}$) based on 16 microsatellite loci

and 2 μ L of template DNA. PCR conditions were as follows: denaturation at 95 °C for 3 min, followed by 35 cycles of 94 °C for 30 s, annealing temperature for 90 s, extension at 72 °C for 60 s, and a final extension for 30 min at 60 °C. PCR products were run on ABI PRISM 3730 automated DNA sequencer with LIZ500 internal size standard (Applied Biosystems) and alleles were scored with GENEMAPPER 4.1 (Fig S1, Supplementary material).

The polymorphic information of 16 microsatellite loci was evaluated using 134 plucked nestlings' feathers of *E. eulophotes* (one per nest). Using MICRO-CHECKER (Oosterhout et al. 2004), no null allele, large allele drop out or stuttering was found (Johansson et al. 2012). The GENEPOP program (Rousset 2008) indicated that all loci conformed to Hardy–Weinberg equilibrium (HWE) after Bonferroni correction except Ae52 which had high polymorphism. The software CERVUS (Kalinowski et al. 2007) showed that the number of alleles per polymorphic locus ranged between 5 and 32 and observed heterozygosity and expected heterozygosity ranged from 0.552 to 0.955 and 0.633 to 0.955, respectively. The polymorphism information content (PIC) values ranged from 0.586 to

0.949 (Table 1). The exclusion probability of each locus for unrelated individuals (P_{ID}) and for individuals with sibling relationship ($P_{ID(sib)}$) were sorted in ascending order, and the combined exclusion probability were 7.02×10^{-19} for P_{ID} and 2.9×10^{-7} for $P_{ID(sib)}$. The $P_{ID(sib)}$ value of any combination of 7 loci among the 16 markers was high enough (>0.99) for individual identification (Fig. 1). Accordingly, CERVUS software was used for identifying individuals with the following criteria: (1) samples were amplified by ≥ 7 loci and ≤ 1 mismatch occurred; (2) samples from a single individual should be identified as identical species and sex, and belong to the same colony in a given year.

In species identification, 61 (75.31 %) of 81 molted feathers were identified as *E. eulophotes* and the remaining 20 (24.69 %) were identified as Little Egret (*Egretta garzetta*) (Table S1, Supplementary material). By sex determination and individual identification, 59 of 61 feather samples of *E. eulophotes* and 18 of 20 feather samples of *E. garzetta*, being successfully amplified by more than 7 loci, were unambiguously assigned to 43 different individuals (32 females and 11 males) and 13 different individuals (9 females and 4 males), respectively (Table S2 and Table S3, Supplementary material). The success rates of molted feathers amplified by ≥ 7 markers was 96.72 % in the current study, in contrast to 47 % of Segelbacher (2002), and the mean amplification success rate of 16 markers was 94.47 % (86.89–98.36 %, Table S4, Supplementary material), which was higher than 33.68 % (15.4–67.7 %) of 17 loci reported by Johansson et al. (2012). These results indicated that our novel multiplex PCR assay was useful for individual identification of molted feathers in *E. eulophotes* and other Ardeids.

Acknowledgments We thank Wei Lei, Site Luo, Leiguan Li and Changfu Mao who helped collect some samples for this study. This research was supported by the National Natural Science Foundation of China (Grant Nos. 41476113, 31000963 and 31272333) and by the Fujian Natural Science Foundation of China (2010Y2007 and 2012J01149).

References

- Dai Y, Zhou X, Fang W, Lin Q, Chen X (2013) Development and cross-species transferability of 23 microsatellite markers from the vulnerable Chinese Egret (*Egretta eulophotes*) using 454 sequencing. *Conserv Genet Resour* 5:1035–1038. doi:10.1007/s12686-013-9962-0
- Hill A, Green MC (2011) Characterization of 12 polymorphic microsatellites for the Reddish Egret, *Egretta rufescens*. *Conserv Genet Resour* 3:13–15. doi:10.1007/s12686-010-9267-5
- Horv  th MB, Mart  nez CB, Negro JJ, Kalm  r L, Godoy JA (2005) An overlooked DNA source for non-invasive genetic analysis in

- birds. *J Avian Biol* 36:84–88. doi:[10.1111/j.0908-8857.2005.03370.x](https://doi.org/10.1111/j.0908-8857.2005.03370.x)
- Huang X, Zhou X, Chen M, Fang W, Chen X (2010) Isolation and characterization of microsatellite loci in vulnerable Chinese Egret (*Egretta eulophotes*: Aves). *Conserv Genet* 11:1211–1214. doi:[10.1007/s10592-009-9921-1](https://doi.org/10.1007/s10592-009-9921-1)
- Huang X, Zhou X, Lin Q, Peng Z, Fang W, Chen X (2012) A novel multiplex PCR assay for species identification in the Chinese Egret (*Egretta eulophotes*) and Little Egret (*E. garzetta*). *Conserv Genet Resour* 4:31–33. doi:[10.1007/s12686-011-9466-8](https://doi.org/10.1007/s12686-011-9466-8)
- Isabel Miño C, Nassif Del Lama S (2009) Molted feathers as a source of DNA for genetic studies in waterbird populations. *Waterbirds* 32:322–329. doi:[10.1675/063.032.0212](https://doi.org/10.1675/063.032.0212)
- IUCN (2014) IUCN red list of threatened species. <http://www.iucnredlist.org/>. Accessed 7 October 2014
- Johansson MP, McMahon BJ, Höglund J, Segelbacher G (2012) Amplification success of multilocus genotypes from feathers found in the field compared with feathers obtained from shot birds. *Ibis* 154:15–20. doi:[10.1111/j.1474-919X.2011.01194.x](https://doi.org/10.1111/j.1474-919X.2011.01194.x)
- Kalinowski ST, Taper ML, Marshall TC (2007) Revising how the computer program CERVUS accommodates genotyping error increases success in paternity assignment. *Mol Ecol* 16:1099–1106. doi:[10.1111/j.1365-294X.2007.03089.x](https://doi.org/10.1111/j.1365-294X.2007.03089.x)
- Oosterhout VC, Hutchinson WF, Wills DP, Shipley P (2004) MICRO-CHECKER: software for identifying and correcting genotyping errors in microsatellite data. *Mol Ecol Notes* 4:535–538. doi:[10.1111/j.1471-8286.2004.00684.x](https://doi.org/10.1111/j.1471-8286.2004.00684.x)
- Rousset F (2008) GENEPOP' 007: a complete re-implementation of the genepop software for Windows and Linux. *Mol Ecol Resour* 8:103–106. doi:[10.1111/j.1471-8286.2007.01931.x](https://doi.org/10.1111/j.1471-8286.2007.01931.x)
- Rudnick JA, Katzner TE, Bragin EA, Rhodes OE, Dewoody JA (2005) Using naturally shed feathers for individual identification, genetic parentage analyses, and population monitoring in an endangered Eastern imperial eagle (*Aquila heliaca*) population from Kazakhstan. *Mol Ecol* 14:2959–2967. doi:[10.1111/j.1365-294X.2005.02641.x](https://doi.org/10.1111/j.1365-294X.2005.02641.x)
- Segelbacher G (2002) Noninvasive genetic analysis in birds: testing reliability of feather samples. *Mol Ecol Notes* 2:367–369. doi:[10.1046/j.1471-8286.2002.00180.x](https://doi.org/10.1046/j.1471-8286.2002.00180.x)
- Shen Z, Qu W, Wang W, Lu Y, Wu Y, Li Z, Zhang C (2010) MPprimer: a program for reliable multiplex PCR primer design. *BMC Bioinform* 11:143. doi:[10.1186/1471-2105-11-143](https://doi.org/10.1186/1471-2105-11-143)
- Wang Z, Zhou X, Lin Q, Fang W, Chen X (2011) New primers for sex identification in the Chinese Egret and other Ardeid species. *Mol Ecol Resour* 11:176–179. doi:[10.1111/j.1755-0998.2010.02879.x](https://doi.org/10.1111/j.1755-0998.2010.02879.x)